BUS63-7

Entrepreneurial Finance

Cap tables, term sheets, and startup financial models — finance from the founder's chair

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The same NPV, WACC, and DCF you already know, pivoted onto the documents that actually decide founder outcomes.

Albert School — 22 hours

Preface

You arrive at this course already fluent in the machinery of corporate finance. From **Financial Accounting & Statements** (BUS13-3) you can read a balance sheet, an income statement, and a cash flow statement, and you know why a profitable company can still run out of cash. From **Financial Diagnosis & Management Control** (BUS23-3) you can compute free cash flow, decompose returns, and build a driver-based budget. From **Corporate Finance** (BUS33-3) you own the time value of money, **NPV**, the cost of capital, and **WACC**. From **Valuation & Investment Analysis** (BUS43-3) you can build a **DCF**, triangulate it against multiples, and stress an assumption until the number moves. And from **Business Models & Value Creation** (BUS23-2) you can decode how a business creates and captures value, and you have already met the **unit economics** — CAC, LTV, gross margin, contribution margin, break-even — that this course will now put to work under real cash pressure.

This course does not replace any of that. It **re-aims** it. Everything you learned was taught from the chair of an analyst pricing a mature, public company, or a CFO allocating capital inside one. Entrepreneurial finance is the same toolkit seen from a different, far more uncomfortable chair: the **founder's**. The founder is the person who has eighteen months of cash, a model built on assumptions nobody has validated, a term sheet on the table whose clauses they only half understand, and a cap table that — one signature at a time — decides how much of their own company they will still own when it is finally worth something.

Three things make finance from this chair genuinely different. First, **cash is the binding constraint, not profit**. A public company optimises returns; a startup optimises **survival**, and the number that kills startups is not negative net income but a bank balance that hits zero. Second, **the financing instruments are unfamiliar and adversarial**. You will not issue a vanilla bond at a quoted yield; you will negotiate a SAFE, a convertible note, or a priced round whose **liquidation preference** and **anti-dilution** clauses can quietly transfer most of an exit to the investor while the headline valuation flatters the founder. Third, **the cap table is the scoreboard**. Valuation, dilution, preferences, and pool top-ups all resolve, eventually, into one table of who owns what — and into one waterfall of who gets paid what when the company is sold.

The discipline this course builds is therefore not a new theory of value. It is the reflex to read every funding decision through two documents at once — the **financial model** that says whether the business works, and the **cap table** that says what the founder keeps if it does — and to quantify, clause by clause and round by round, the gap between the valuation a founder celebrates and the outcome a founder actually receives.

Contents

1 The financial lifecycle of a startup

1.1 Two companies, same revenue, opposite health

Two startups close the year at €2M of annual revenue, both growing 100% year on year. The first burned €4M of cash to get there and has €1M left in the bank; the second burned €500k and has €3M left. On a strategy slide they look identical — same revenue, same growth. On the only dimension that decides whether they are alive next year, they are opposites: the first has roughly three months of life left, the second has years. Mature- company finance would rank them on margin and return; startup finance ranks them on **how long until the cash runs out**, and that question reorganises everything.

A startup is not a small large-company. It is a **temporary organisation searching for a repeatable, scalable business model**, and it is doing that search while spending money it has not yet earned. Its finances are therefore organised around a single arc: raise cash, spend it to buy evidence that the model works, and convert that evidence into the next, larger raise — until, if all goes well, the business generates enough cash to fund itself.

1.2 The valley of death and the funding staircase

Definition 1 (the startup cash lifecycle) A startup's cumulative cash position typically follows a **J-curve**. Early spending on product and go-to-market runs ahead of revenue, so cumulative cash falls into a trough — the **valley of death** — before (if the model works) revenue overtakes spending and the curve turns up toward **cash-flow break-even**, the point at which the company no longer needs external money to survive.

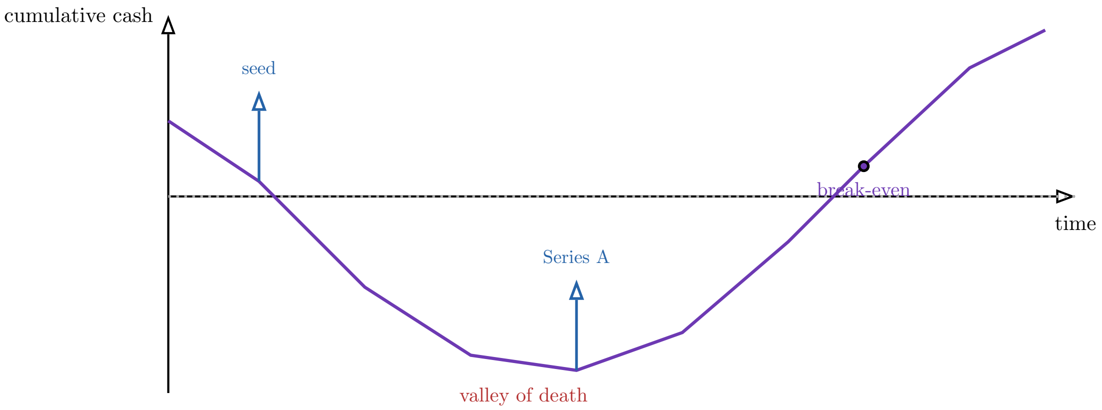

Figure 1: The startup cash J-curve. Cumulative cash falls as spending outruns revenue, bottoms out in the *valley of death*, and — only if the model works — climbs back to *break-even*. Each funding round is a vertical injection that resets the runway and buys time to reach the next milestone. The arc, not the annual profit, is the object of startup finance.

The staircase of rounds — **pre-seed**, **seed**, **Series A**, **Series B**, and beyond — is not arbitrary. Each round buys the evidence that unlocks the next. The unit of progress is the **milestone**, and milestones change qualitatively by stage.

Stage	Central question	What money buys	Revenue pattern
Pre-seed / seed	Does anyone want this?	Product + first customers	€0 → first revenue, erratic
Series A	Is the model repeatable?	A working go-to-market engine	Growing, still unprofitable
Series B+	Does it scale?	Scale the proven engine	Large, accelerating
Growth / pre-exit	Can it stand alone?	Market leadership, path to profit	Approaching break-even

Table 1: The funding staircase. Each stage answers a different question, and investors price the round on whether the previous question has been convincingly answered. Cash needs rise at every step; the revenue pattern is the evidence that justifies the next raise.

Definition 2 (financial inflection points) The moments at which a startup's financial character changes qualitatively:

- **First revenue** — the model stops being a hypothesis and starts producing data.
- **Product–market fit** — retention and organic pull appear; growth stops being bought one customer at a time. This is what a Series A pays for.
- **Unit-economic positivity** — each customer becomes worth more than it costs to acquire; growth starts funding itself rather than consuming cash.

- **Cash-flow break-even** — operating cash flow turns non-negative; external financing becomes a choice rather than a survival necessity.

Common pitfall The biggest conceptual error carried over from corporate finance is treating **net income** as the health metric. For a startup, the binding constraint is the **bank balance**. A company can be deeply unprofitable for years and perfectly healthy (well-funded, on milestone), or modestly unprofitable and three weeks from death. The lifecycle is read on the **cash** curve, not the P&L.

2 Building a startup financial model

2.1 Why a top-down number is a lie

A founder pitches: "the market is €10 billion; we will capture just 1% in three years — €100M of revenue." This is a **top-down** model, and investors discount it to zero on sight, because "1% of a big number" is a wish, not a plan: it names no customer, no salesperson, no conversion rate, nothing that can be tested or managed. The credible alternative builds the same revenue from the bottom up — from the **drivers a founder actually controls** — and in doing so exposes exactly which assumption the whole model rests on.

Definition 3 (bottoms-up driver model) A revenue projection built by **multiplying together the operational drivers that produce a sale**, rather than by taking a share of a market. For example:

$$\text{Revenue} = \text{leads} \times \text{conversion rate} \times \text{average price} \times \text{purchase frequency}.$$

Each driver is a number the founder can forecast from real data and a sales team can be held to. The discipline is not the formula but the **traceability**: every euro of projected revenue is the product of named, defensible assumptions.

Worked example — a bottoms-up monthly build. A B2B SaaS startup models new revenue from its sales motion:

- 2 sales reps, each generating 40 qualified leads/month → 80 leads/month
- 20% close rate → 16 new customers/month
- €500/month average contract → €8,000 of **new** MRR added each month

MRR (monthly recurring revenue) is then rolled forward with churn: next month's MRR = this month's MRR × (1 − monthly churn) + new MRR. At 3% monthly churn, MRR grows toward a ceiling where new revenue just replaces churned revenue — a ceiling the founder can compute **before** hiring, and raise by adding reps, lifting the close rate, or cutting churn.

2.2 From drivers to three coherent statements

A startup model is not one spreadsheet tab but **three linked statements plus a cap table**, and the discipline — exactly as in BUS13-3 — is that they must **reconcile**. Revenue and costs build the **P&L**; the P&L plus the timing of cash in and out builds the **monthly cash flow**; the cash flow drives the **bank balance**; and the financing that refills that balance must match the **cap table**. A model whose three statements do not tie is not conservative or aggressive — it is simply wrong.

Definition 4 (the four linked components)

1. **P&L (income statement)** — revenue minus costs; shows profitability over a period.
2. **Cash flow statement** — actual cash in and out, **monthly**, capturing timing differences (a signed annual contract is revenue spread over 12 months but cash that may arrive now or in arrears).
3. **Bank-balance / runway schedule** — the running cash position; the line that hits zero.
4. **Cap table** — who owns what; updated whenever a financing round refills the bank balance.

Common pitfall **P&L revenue is not cash.** The single most common modelling error is forecasting a healthy, growing P&L while ignoring **when the cash actually moves**. A company that bills annually in advance is cash-rich relative to its P&L; one that bills in arrears (net-60 enterprise terms) can show a profit on the P&L and still go insolvent waiting to be paid. Startups model **monthly cash**, not annual accruals, precisely because timing is where they die.

2.3 Headcount: the dominant cost driver

For most startups, **people are 60–80% of total spend**. Forecasting cost therefore means forecasting **headcount** — a hiring plan with start dates, fully-loaded salaries (gross pay plus employer charges, which in France add roughly 40–45% on top of gross), and the ramp time before a new hire is productive. Because hiring is the biggest lever a founder pulls, it is also the biggest way they kill the company (Chapter 8).

> **Worked example — headcount and the monthly cash flow.** Plan: 10 staff at an average **fully-loaded** €6,500/month → €65,000/month payroll. Add €20,000/month of other operating costs (tools, rent, marketing) → €85,000/month of cash out. Against €40,000/month of cash revenue, the company spends €45,000 of **net** cash each month. With €900,000 in the bank, that is exactly 20 months of runway — **before** any new hire. Adding two engineers at €6,500 raises net burn to €58,000 and cuts runway to 15.5 months. The model makes that trade-off visible in advance, which is its entire point.

The horizon for a startup model is typically **18 to 24 months** — long enough to reach the next milestone and raise again, short enough that the assumptions are not pure fiction. The output a founder defends is not a single base case but the **monthly path of the bank balance**, with the date it would hit zero clearly marked.

3 Unit economics: does each customer make money?

3.1 A growing business that loses money on every sale

A delivery startup triples its orders every quarter; the growth chart is vertical and investors are excited. Then someone computes the economics of a **single** order: the company collects €18, pays the courier €12, the restaurant €9, and €2 in payment and support costs — losing €5 on every order it is so proud of growing. Faster growth here means faster losses. **Unit economics** is the discipline of zooming in from the company to the single customer or single transaction and asking the question the growth chart hides: **does this unit make money, and how much?**

You met these metrics in **Business Models & Value Creation** (BUS23-2), where they were defined once. Here they are mobilised under cash pressure, with the thresholds that decide whether a startup is fundable.

3.2 The core metrics

Definition 5 (contribution margin and gross margin)

- **Gross margin** = (revenue − cost of goods sold) ÷ revenue. The share of each euro of revenue left after the **direct** cost of delivering the product. Software: 70–90%. Marketplaces and hardware: often 10–40%.
- **Contribution margin** = revenue − **all variable costs** of serving one more unit (COGS plus variable selling, payment, support). What one incremental sale contributes toward covering fixed costs and, eventually, profit.

Definition 6 (CAC, LTV, payback)

- **CAC** (customer acquisition cost) = total sales & marketing spend ÷ new customers acquired in the same period. What it costs to buy one customer.
- **LTV** (lifetime value) = the total **gross-margin contribution** a customer delivers over their whole relationship. For a subscription with monthly contribution m and monthly churn c, average lifetime is $1/c$ months, so

$$\text{LTV} = m \times \frac{1}{c} = \frac{\text{ARPU} \times \text{gross margin}}{c}.$$

- **Payback period** = CAC ÷ monthly contribution per customer. How many months of a customer's margin it takes to earn back the cost of acquiring them.

Worked example — a SaaS unit-economics teardown. A SaaS product charges €100/month with €20/month of hosting and support cost.

- Contribution = €100 − €20 = €80/month; gross margin = 80%.
- Monthly churn 3% → average lifetime = 1/0.03 ≈ 33 months.
- **LTV** = €80 × 33 ≈ **€2,640**.
- **CAC** = €1,200 (€60,000 of marketing ÷ 50 new customers).
- **LTV/CAC** = 2,640 ÷ 1,200 ≈ **2.2×**.
- **Payback** = 1,200 ÷ 80 = **15 months**.

The verdict: each customer is profitable, but the 2.2× ratio is **below the 3× rule of thumb** and 15-month payback is long. This business is viable but not yet fundable-for-scale; the levers are obvious — cut churn (raises lifetime), cut CAC, or raise price.

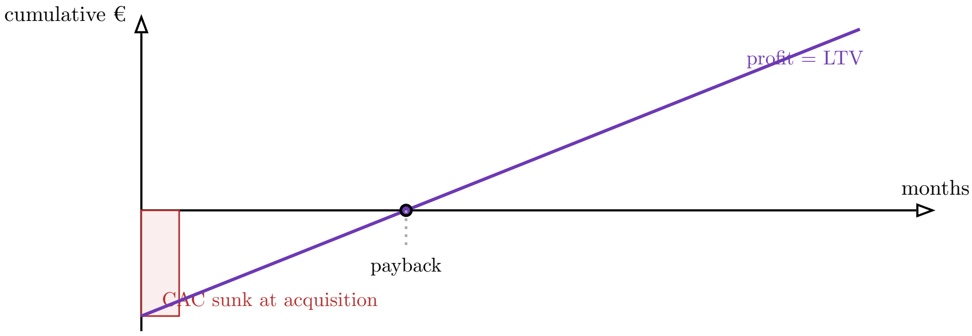

Figure 2: The payback curve for one customer. The acquisition cost (CAC) is sunk on day one; each month of contribution margin climbs the cumulative line back toward zero. The crossing point is the *payback period* everything earned afterwards, out to the end of the customer's life, is the profit that LTV measures. A short payback means cash returns quickly to fund the next acquisition — which is why payback, not just LTV/CAC, governs how fast a startup can grow on its own cash.

Definition 7 (the viability thresholds) Rules of thumb investors apply (sector-dependent, not laws):

- **LTV/CAC ≥ 3×** — below this, acquisition economics are too thin to build a business on.
- **Payback ≤ 12 months** (consumer) or **≤ 18–24 months** (enterprise) — longer payback ties up cash and demands more financing to grow.
- **Gross margin** high enough that the model can ever carry its fixed costs — the structural ceiling on everything else.

Common pitfall **LTV is only as honest as its churn and margin inputs.** Two abuses are routine: using **revenue** instead of **gross-margin contribution** (inflating LTV by the cost of delivery), and assuming a low churn the company has not lived long enough to observe (a 2-year-old startup cannot know its 5-year retention). A high LTV/CAC built on optimistic churn is the most common way a model flatters a business that does not work.

4 Cash dynamics: burn, runway, and break-even

4.1 The only deadline that cannot be moved

A founder can renegotiate a contract, delay a launch, or talk an investor into more time on almost anything — except one date. When the bank balance reaches zero and payroll is due, the company stops. **Burn** and **runway** are the two numbers that name that date, and a founder who cannot recite them from memory is not in control of the company.

Definition 8 (gross burn, net burn, runway)

- **Gross burn** = total cash **out** per month (all operating cash costs). What the company spends regardless of what it earns.
- **Net burn** = gross burn − cash revenue = the **net** monthly drain on the bank balance. This is the number that depletes runway.
- **Runway** = cash in bank ÷ net burn = the number of months until the balance hits zero **at the current net burn**.

> **Worked example — computing the deadline.** A startup holds €1,200,000. It spends €250,000/month (gross burn) and collects €100,000/month of cash revenue. Net burn = €150,000/month. **Runway** = 1,200,000 ÷ 150,000 = **8 months**. If revenue grows to €175,000/ month while costs hold, net burn falls to €75,000 and runway doubles to 16 months — the same cash, twice the life, bought entirely by growing revenue against a flat cost base.

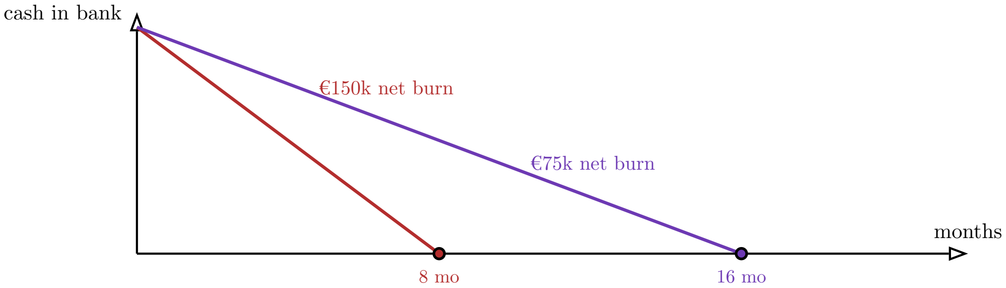

Figure 3: Runway is the slope of the cash-depletion line. Halving net burn — by growing revenue or cutting cost — does not change the starting cash but doubles the time to zero. Every founder decision (a hire, a price change, a marketing push) is, in cash terms, a change to the slope of this line.

4.2 The levers — and the one that kills growth

Runway can be extended two ways: spend less, or earn more. But not every extension is wise. Cutting the marketing that drives growth, or the engineers building the product, can lengthen runway while **destroying the very milestone the runway exists to reach** — a company that survives 24 months but has nothing to show at month 24 is no better off than one that died at month 12.

Definition 9 (runway levers)

- **Cut burn** — reduce headcount, trim discretionary spend, renegotiate contracts. Fast and certain, but can amputate growth.
- **Grow revenue** — accelerate sales, raise prices, improve retention. The best lever, because it extends runway **and** advances the milestone, but the slowest to act.
- **Improve cash timing** — bill annually upfront instead of monthly, collect receivables faster, stretch payables. Frees cash without touching the P&L.
- **Raise more** — the last resort, and the most expensive in dilution (Chapter 7).

The art is extending runway **without killing growth**: cut the spend that does not advance the milestone, protect the spend that does.

4.3 Break-even

Definition 10 (break-even point) The level of activity at which **total contribution margin exactly covers fixed costs**, so profit is zero. In units,

$$\text{break-even volume} = \frac{\text{fixed costs}}{\text{contribution margin per unit}};$$

in revenue,

$$\text{break-even revenue} = \frac{\text{fixed costs}}{\text{contribution-margin ratio}}.$$

Worked example — moving the break-even point. A startup has €120,000/month of fixed costs and an 80% contribution-margin ratio. Break-even revenue = 120,000 ÷ 0.80 = **€150,000/month**. At a €100 price that is 1,500 customers. Now change one assumption at a time:

- **Raise price 10%** (margin ratio rises toward ≈ 82%) → break-even falls to ≈ €146k.
- **Variable costs rise** so the ratio drops to 70% → break-even jumps to €171k — a 14% cost swing moves the target by 14%.
- **Hire ahead of revenue**, lifting fixed costs to €160k → break-even rises to €200k, demanding 33% more customers just to stand still.

Break-even is not one number but a **surface** that pricing, volume, and cost assumptions push around — which is exactly why founders must know which lever moves it most.

5 The investor ecosystem and how VC funds make money

5.1 Why the source of the money changes the deal

Two founders each raise €2M. The first takes it from a single business angel who wrote a personal cheque and wants the company to be a good business. The second takes it from a venture fund whose own investors expect the fund to **triple**, and who therefore needs this particular company to have a shot at returning the **entire fund** by itself. Same €2M, utterly different expectations about growth, exit, and how hard the company will be pushed. To negotiate with an investor, a founder must understand **where that investor's money comes from and what return it is promised** — because that, not the cheque size, determines what they will demand at the table.

5.2 The funding sources

Definition 11 (the early-stage funding landscape)

- **Bootstrapping** — funding growth from revenue and founders' own money. No dilution, total control, but growth capped at what cash flow allows.
- **Business angels** — wealthy individuals investing their own money, €10k–€500k, often the first outside cheque. **Individual** angels invest personally; **institutional** angel groups or syndicates pool several angels. They add judgement and contacts, not just cash.
- **Accelerators** (Y Combinator, Techstars, Station F programmes) — small cheque + intensive programme + network, in exchange for a few percent of equity. They buy access and signalling as much as money.
- **Seed funds** — small VC funds specialised in the first institutional round.
- **Venture capital funds** — the institutional engine of the next sections, financing Series A and beyond.

5.3 How a VC fund actually works

A venture fund is itself a financial structure with its own investors, incentives, and clock — and every one of those flows down into the term sheet a founder is handed.

Definition 12 (LPs, GPs, and the fund structure)

- **Limited Partners (LPs)** — the fund's investors: pension funds, endowments, family offices, funds-of-funds. They supply the capital and bear the risk, but take no role in day-to-day decisions (hence **limited**).
- **General Partners (GPs)** — the venture firm that raises the fund, selects investments, sits on boards, and manages the portfolio.
- The fund is a **closed-end vehicle**, typically **10 years**: capital is called and invested over the first 3–5 years, then harvested as portfolio companies exit.

Definition 13 (the 2-and-20 economics) GPs are paid two ways:

- **Management fee** — **2% per year** of committed capital, to run the firm.
- **Carried interest ("carry")** — **20% of the fund's profits**, usually above a **hurdle** return owed to LPs first. Carry is where GPs make real money, and it is why GPs need **large** exits, not merely positive ones.

Definition 14 (vintage year) The **vintage** is the year a fund starts investing. Returns are benchmarked within a vintage cohort because the macro environment (valuations, exit windows) at the time of investing dominates outcomes — a great fund in a bad vintage may underperform a mediocre fund in a great one.

5.4 The power law — why VCs behave as they do

Definition 15 (the venture power law) Venture returns are not normally distributed; they follow a **power law**. In a typical fund, **most investments return little or nothing, and a tiny number return the bulk of the fund**. A GP therefore does not seek companies that will **probably** do **fine** — a 2× outcome that cannot move a fund of dozens of investments. They seek the few companies with a **real chance of a 50–100× outcome**, even at a high risk of total loss.

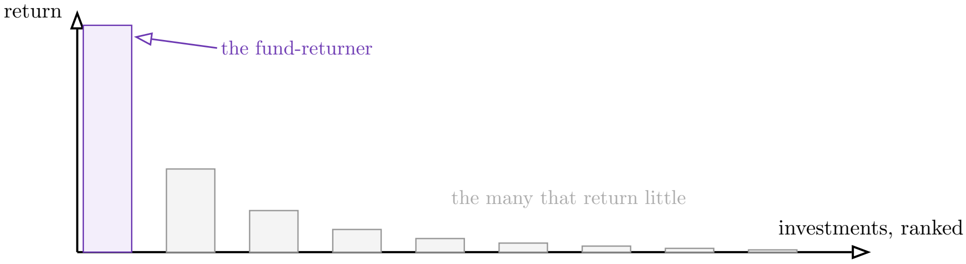

Figure 4: The venture power law. One investment (the *fund-returner*) produces more value than all the others combined; the long tail returns little or nothing. This single fact explains why a VC pushes portfolio companies for outsized growth, why "a solid, profitable small business" is the wrong pitch for venture money, and why investors accept high failure rates in exchange for access to the rare extreme winner.

5.5 Stage specialisation and private equity

Definition 16 (VC stage and type specialisation)

- **Early-stage VC** — seed / Series A; high failure rate, smallest cheques, largest multiples on the winners.
- **Growth-stage VC** — later rounds into proven companies; lower risk, lower multiples, larger cheques.
- **Corporate VC (CVC)** — a fund run by a large company (Google Ventures, etc.) seeking strategic fit as well as financial return.

Definition 17 (private equity vs venture capital) **Private equity** buys **established, cash-generating** companies, usually with **debt** (a **leveraged buyout**), to improve and resell. It splits into:

- **Buyout** — acquire control of mature companies, often using leverage.
- **Growth equity** — large minority stakes in fast-growing but established firms, the bridge between venture and buyout.

The contrast with venture is structural: VC buys **minority** stakes in **unprofitable, high-growth** companies funded by **equity**; PE buys **control** of **profitable** companies funded substantially by **debt**.

Remark "Select investors based on criteria beyond the cheque." Two term sheets at the same valuation are not equal: the investor's **stage fit**, **portfolio** (do they back competitors?), **reserves** for follow-on rounds, **board behaviour**, and **reputation with the next round's investors** often matter more than a 10% difference in price. The cheapest money is frequently the most expensive partner.

6 Term sheets clause by clause

6.1 Why the valuation is not the deal

A founder receives two term sheets. Term sheet A values the company at €10M; term sheet B values it at €12M. The founder, naturally, leans toward B — until the clauses are read. B carries a **2× participating liquidation preference** and **full-ratchet anti-dilution**; A carries a **1× non-participating** preference and **broad-based weighted-average** anti-dilution. On any realistic exit, the founder **keeps more money under the lower-valuation term sheet A**. The headline number is the part of the deal designed to be seen; the **clauses** are where the economics actually live, and reading them is the central skill of this chapter.

Definition 18 (term sheet) A short, mostly non-binding document setting out the **key terms** of an investment before the long legal contracts are drafted. Its clauses fall into two families: **economic** (who gets how much money, and when) and **control** (who decides what). A founder must be able to quantify the economic clauses and assess the control clauses.

6.2 The instruments: SAFE, convertible note, priced round

Definition 19 (the three ways to take early money)

- **Priced equity round** — investors buy shares at an agreed price, set by an agreed **valuation**. Clean and final, but requires negotiating a valuation now and the most legal work.
- **Convertible note** — a **loan** that converts into equity at the next priced round, carrying **interest** and usually a **discount** and/or **valuation cap**. Defers the valuation debate; it is debt until it converts.
- **SAFE** (Simple Agreement for Future Equity) — like a convertible note but **not debt**: no interest, no maturity. Converts at the next round, typically on a **cap** and/or **discount**. Fast, cheap, founder-friendly — now the dominant early-stage instrument.

Definition 20 (discount and valuation cap) Convertibles and SAFEs reward early investors for taking early risk via:

- **Discount** — the early investor converts at, say, 20% below the next round's price.
- **Valuation cap** — a ceiling on the valuation at which they convert, regardless of the next round's headline. The cap protects the early investor when the company does very well.

The investor converts at **whichever gives them more shares** — the lower of the cap-implied price and the discounted price.

Worked example — a post-money SAFE converting. An angel puts €500,000 into a **post-money SAFE** with a **€5M cap**. Eighteen months later the Series A prices the company at €10M pre-money. The SAFE converts at the **cap**, not the round price: €500,000 ÷ €5M = **10% of the company** (on a post-money-SAFE basis, fixed before the new round's dilution). Had the company done less well and the round priced **below** the €5M cap, the SAFE would instead convert at the round price (with any discount applied). The cap is what lets a small early cheque become a meaningful stake when the company succeeds.

Common pitfall **SAFEs are not free of dilution — they are deferred dilution.** Because a SAFE postpones the valuation, founders often stack several SAFEs at different caps and lose track of how much of the company they have already sold. When the priced round finally arrives, **all** the SAFEs convert at once, and founders routinely discover they own far less than they thought. Always model the **as-converted** cap table before signing the next SAFE, not after.

6.3 Liquidation preference — the most important economic clause

Definition 21 (liquidation preference) The right of preferred shareholders to be paid **before** common shareholders (founders, employees) when the company is sold or wound up. Two dimensions:

- **Multiple** — **1×** (get your money back first), **2×**, **3×** (get a multiple of it first).
- **Participation**:
  - **Non-participating** — the investor takes **either** the preference **or** their as-converted ownership share, **whichever is greater** — not both.
  - **Participating** ("double dip") — the investor takes the preference **first**, **then also** shares pro-rata in what remains, as if they had converted.

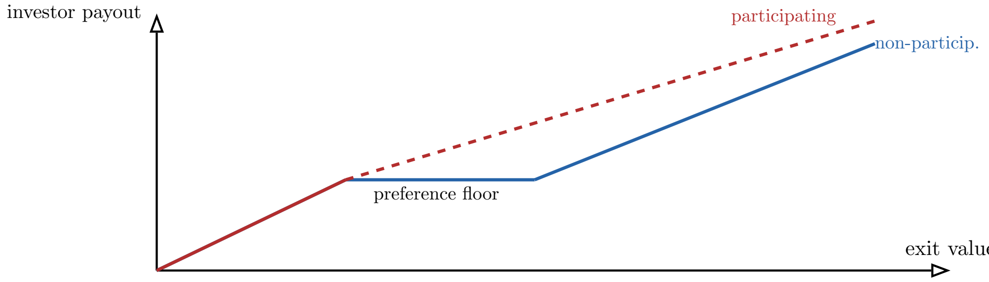

Figure 5: Investor payout vs exit value. With a *non-participating* preference the investor takes the preference at low exits, then *converts* to common once their ownership share exceeds it — a single, capped-then-pro-rata payoff. With *participating* preferred the investor takes the preference *and* shares the rest, sitting permanently above the non-participating line. The gap between the two lines is value transferred directly from founders to the investor.

6.4 Anti-dilution — protection against a down round

Definition 22 (anti-dilution protection) Adjusts the price at which earlier preferred shares convert if the company later raises at a **lower** price (a **down round**), issuing the earlier investor extra shares to compensate. Two flavours:

- **Full ratchet** — the earlier investor is repriced as if they had paid the **new, lower** price on **all** their shares. Maximally punitive to founders.
- **Broad-based weighted average** — the conversion price is adjusted **partially**, weighted by the **size** of the down round relative to the whole cap table. Far gentler, and the market standard.

Worked example — full ratchet vs weighted average. An investor bought at €2.00/share. A down round prices at €1.00/share.

- **Full ratchet**: their conversion price drops to €1.00 — they effectively double their share count, diluting founders heavily, no matter how small the down round.
- **Broad-based weighted average**: the conversion price falls only part-way (say to €1.80), because the formula weights the small down round against the large existing share base. The founder gives up a sliver, not a slab.

On the same down round, full ratchet can cost founders several times more ownership than weighted average — which is why this one word in the term sheet is worth negotiating hard.

6.5 Governance and exit clauses

Definition 23 (the control and exit clauses)

- **Drag-along** — if a defined majority agrees to sell the company, they can **force** minority holders (including founders) to sell too, preventing a holdout from blocking an exit.
- **Tag-along (co-sale)** — if a major shareholder sells, minority holders may **join** the sale on the same terms, so they are not left behind.
- **Right of first refusal (ROFR)** — the company or investors may **match** any offer before shares are sold to an outsider.
- **Pro-rata rights** — an investor's right to invest in **future** rounds to **maintain their percentage** — valuable to investors, and a future dilution source founders must track.

Common pitfall **Control clauses bite when things go wrong, not when they go right.** In the good scenario everyone is aligned and the clauses are invisible. Their entire purpose is to govern the **bad** and **contested** scenarios — a forced sale, a blocked financing, a founder departure. Founders skim them because they are imagining success; they are written by investors who are imagining failure. Read them as if the company is in trouble, because that is when they fire.

7 Cap tables and dilution across rounds

7.1 One signature, a permanent transfer of ownership

A founder owns 100% of their company on the day they incorporate it. By the day it is sold, they may own 20%, or 8%, or — through preferences — effectively less. Nothing was stolen; each slice was sold, knowingly, in exchange for the cash that let the company grow. The **cap table** (capitalisation table) is the running ledger of that bargain, and **dilution** is its arithmetic. A founder who cannot model their own cap table is negotiating blind.

7.2 Pre-money, post-money, and price per share

Definition 24 (pre-money and post-money)

- **Pre-money valuation** — what the company is agreed to be worth **before** the new money.
- **Post-money valuation** = pre-money + amount raised — the value **after** the cash is in.
- The new investor's stake is

$$\text{investor \%} = \frac{\text{amount raised}}{\text{post-money valuation}}, \quad \text{price per share} = \frac{\text{pre-money}}{\text{existing shares}}.$$

Worked example — the mechanics of one round. A company has 8,000,000 shares and raises **€1M at a €4M pre-money** valuation.

- Post-money = 4 + 1 = **€5M**.
- Investor stake = 1 ÷ 5 = **20%**.
- Existing holders keep 80%, so total shares = 8,000,000 ÷ 0.80 = **10,000,000**; the investor receives 2,000,000 new shares.
- Price per share = €4M ÷ 8,000,000 = **€0.50** (check: 2,000,000 × €0.50 = €1M ✓).

Common pitfall **Pre-money vs post-money is where founders are quietly diluted.** "I'll invest €1M at a €5M valuation" is ambiguous: at €5M **pre** the investor gets $1/6 \approx 16.7\%$; at €5M **post** they get $1/5 = 20\%$. On a €1M cheque that ambiguity is over 3% of the company. **Always confirm which one is meant** — and remember that any **option pool** created "pre-money" is carved out of the **founders'** stake before the investor's percentage is computed (the "option-pool shuffle").

7.3 Dilution across successive rounds

Dilution **compounds**: each round multiplies the founders' stake by (1 − the new investors' share). The clearest way to see it is to follow one cap table through three rounds.

Worked example — founder dilution from seed to Series B. Founders start with 8,000,000 shares = 100%. Three rounds follow, each selling a fresh slice:

Round	New %	Raised	Post-money	Price/sh	Founders %
Founding	—	—	—	—	100%
Seed	20%	€1M	€5M	€0.50	80%
Series A	25%	€5M	€20M	€1.50	60%
Series B	20%	€15M	€75M	€4.50	48%

Each round multiplies the founders' stake: $100\% \times 0.80 \times 0.75 \times 0.80 = 48\%$. The final fully-diluted cap table:

Holder	Shares	Ownership
Founders	8,000,000	48%
Seed investors	2,000,000	12%
Series A investors	3,333,333	20%
Series B investors	3,333,333	20%
Total	**16,666,667**	**100%**

The founders' **percentage** fell by more than half, yet the **value** of their stake rose from nothing to 48% of €75M ≈ €36M. Dilution is not loss if each round raises the whole pie by more than it shrinks the founders' slice — which is the entire bet of venture financing.

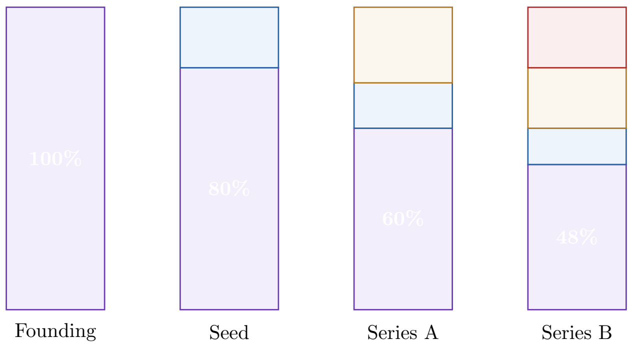

Figure 6: Cap-table evolution. Each bar is the whole company; the bottom (purple) band is the founders' shrinking share — 100% → 80% → 60% → 48% — as seed, Series A, and Series B investors are stacked on top. The founders' *slice* shrinks every round, but the *bar* (the company's value) grows far faster, which is why a smaller percentage of a larger company can still make founders far wealthier.

Remark **BSPCEs** (*bons de souscription de parts de créateur d'entreprise*) are the French employee stock-option instrument — the equivalent of an ESOP elsewhere. A pool of shares (often 10–15%) is reserved to hire and retain employees. It sits on the cap table like any other holder and dilutes the founders; crucially, when an investor demands a pool be created or topped up "pre-money," that dilution falls on the **founders and existing holders**, not the incoming investor.

8 Modelling the exit: the liquidation waterfall

8.1 Why the same exit pays founders differently

The company from Chapter 7 is sold. The price determines the size of the pie, but it does **not** directly determine what the founders receive — the **liquidation waterfall** does. The waterfall is the ordered procedure by which the sale proceeds are distributed: **preferences first, common last**. At a high enough exit everyone simply shares pro-rata; at a low exit the preference stack can absorb **everything**, leaving founders and employees with nothing despite a headline sale price that sounds like success.

Definition 25 (the liquidation waterfall) The sequence in which exit proceeds are paid:

1. **Liquidation preferences** to preferred investors (in stacked order, usually last-in-first-out).
2. **Remaining proceeds** to common holders (founders, employees) — and to any **participating** preferred, and to any **non-participating** preferred who **converted** because their as-converted share beat their preference.

Each non-participating investor independently takes **the greater of** their preference or their as-converted ownership of the proceeds.

8.2 A worked waterfall

Worked example — the same cap table at four exit values. Use the Chapter 7 cap table. Investors invested €1M (seed), €5M (A), €15M (B) — a **€21M** total preference stack, all **1× non-participating**. Ownership: Founders 48%, Seed 12%, A 20%, B 20%.

Computing the waterfall at four exits (each non-participating holder takes the greater of preference or as-converted share, resolved consistently):

Exit value	Founders	Seed	Series A	Series B
€21M (≈ money back)	€0.0M	€1.0M	€5.0M	€15.0M
€75M (1× last post)	€36.0M	€9.0M	€15.0M	€15.0M
€225M (3×)	€108.0M	€27.0M	€45.0M	€45.0M
€750M (10×)	€360.0M	€90.0M	€150.0M	€150.0M

Read the top row carefully: at a €21M exit — a sale that **returns investors their money** — the founders receive **nothing**, because the €21M preference stack consumes the entire price before common holders are paid. At €75M and above, every investor's as-converted share (e.g. B's 20% × €75M = €15M) meets or beats their preference, so they convert and the proceeds split purely by ownership — founders take their full 48%. The preference protects investors on the downside and **costs them nothing** on the upside; it is a one-way option written by the founders.

Worked example — the cost of a participating preference. Now suppose Series B's €15M preference were **1× participating** instead of non-participating, at the €75M exit. B first takes its €15M off the top; the remaining €60M is then split pro-rata among **everyone, including B**:

- B: €15M + 20% × €60M = €15M + €12M = **€27M** (versus €15M non-participating).
- Founders: 48% × €60M = **€28.8M** (versus €36M).

- Seed: 12% × €60M = €7.2M; Series A: 20% × €60M = €12M.

The single word **participating** moved €7.2M from the founders to Series B at the **same** exit price. Multiply that across a stack of participating rounds and the founders' share of a "successful" exit can be a fraction of their headline ownership.

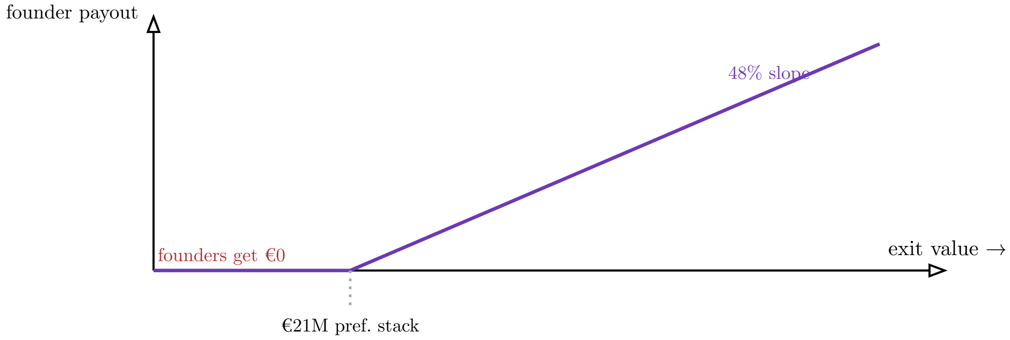

Figure 7: The founders' payoff against exit value. Below the €21M preference stack the founders receive *nothing* — the flat segment — because investors are paid first. Only above the stack do founders participate, at the slope of their 48% ownership. The kink is the whole reason a founder must model the waterfall: it locates the exit value below which their equity is worthless, however large the cap table says their stake is.

Remark **Cap-table structure shapes incentives at exit.** A heavy preference stack with participation makes founders and employees indifferent to — or even hostile toward — a modest exit that pays investors well but leaves common holders with little, while investors with a guaranteed preference may prefer a safe small sale to a risky larger one. Misaligned waterfalls are a frequent cause of fights over whether and when to sell. The clean term sheet (1× non-participating) keeps everyone pulling toward the same outcome: the biggest possible pie.

9 Alternative financing: debt and revenue-based instruments

9.1 When selling equity is the wrong tool

A profitable-at-the-unit-level startup needs €1M to buy inventory it will sell in four months. Raising an equity round to fund a four-month working-capital cycle would mean **permanently** selling a slice of the company to cover a **temporary** need — the most expensive money there is for the job. Equity is priced for the risk of total loss; when a financing need is **lower-risk and self-liquidating**, cheaper, non-dilutive instruments fit better. The founder's task is to **match the instrument to the cash-flow profile**.

Definition 26 (venture debt) A **loan** to a venture-backed startup, usually **alongside or just after an equity round**, from specialist lenders. It carries **interest** (often 10–14%) plus **warrants** (a small equity kicker — the right to buy shares, giving the lender upside) and sometimes fees. It is **minimally dilutive** and **extends runway** between rounds, but it must be **repaid on a schedule** regardless of performance — and it adds a creditor who sits **ahead of everyone** in a liquidation.

Definition 27 (revenue-based financing (RBF)) Capital advanced in exchange for a **fixed percentage of future revenue** until a **capped total** is repaid (e.g. repay 1.5× the advance). No equity, no fixed schedule: repayments **flex with revenue** — faster when sales are strong, slower when weak. It suits businesses with **predictable, recurring revenue** (SaaS, e-commerce) and **no hard asset to pledge**.

9.2 Computing the effective cost

The headline of an alternative instrument hides its true cost, which depends on **how fast** it is repaid.

Worked example — the effective cost of RBF. An RBF provider advances **€1,000,000** and is repaid **€1,300,000** (a 1.3× cap) out of 10% of monthly revenue. The €300,000 fee is fixed, but its **annualised** cost depends entirely on speed:

- Repaid in **12 months** → 30% effective annual cost.
- Repaid in **24 months** → roughly **half** that, 14% — the same euros spread over twice the time.

The faster the company grows (the quicker it repays), the **more expensive** RBF becomes in annualised terms — the opposite intuition to most debt. RBF is cheap insurance for slow-but-steady revenue and a costly tax on a rocket.

Worked example — the all-in cost of venture debt. A lender provides **€2M** at **12%** interest plus **warrants** for 1% of the company. The interest is the visible cost; the warrants are the hidden one — if the company later exits for €200M, that 1% is worth €2M, **doubling** the economic cost of the loan well beyond its 12% coupon. Venture debt looks cheap on the term sheet and can be expensive at exit, exactly when it is least noticed.

Definition 28 (matching the instrument to the profile)

- **Equity** — for high-risk, pre-revenue, or deeply unprofitable phases where no lender will lend and the company cannot promise repayment.
- **Venture debt** — to extend runway **between equity rounds** for a company with a credible path to the next raise, accepting a fixed repayment schedule.
- **Revenue-based financing** — to fund **growth** (marketing, inventory) for a company with **predictable recurring revenue** and good unit economics, where repayment can flex with sales.

The wrong match — equity for a working-capital need, or debt for a pre-revenue gamble — is one of the costliest financial errors a founder makes.

10 Common founder financial mistakes

10.1 The mistakes are predictable — and therefore avoidable

Founders fail in financially **patterned** ways. The same handful of errors recur across companies and sectors, which means they can be named in advance and designed around. Each of the four below is the shadow side of a competence built earlier in this book.

Definition 29 (the four recurring mistakes)

1. **Over-hiring ahead of revenue.** Raising a round and immediately scaling headcount on the **assumption** that revenue will follow. Because people are 60–80% of burn (Chapter 2), a hiring plan built on unproven revenue converts a long runway into a short one and forces a premature, weak-position raise — or layoffs. **Avoidance:** hire **behind** validated demand, not ahead of projected demand.
2. **Mispricing.** Setting price by cost-plus or by fear rather than by value, and then leaving it untouched. Price flows directly into contribution margin, LTV, payback, and break-even (Chapters 3–4), so a pricing error contaminates every unit-economic metric at once. **Avoidance:** treat price as the highest-leverage variable in the model and test it deliberately.
3. **Ignoring unit economics.** Scaling a business that loses money on every transaction, trusting that volume will fix it. If the contribution margin is negative, **growth makes the losses larger** (Chapter 3). **Avoidance:** prove positive unit economics **before** pouring cash into acquisition.
4. **Conflating revenue growth with cash health.** Reading a rising top line as proof the company is healthy, while the bank balance falls. Revenue is a P&L number; survival is a cash number, and the two diverge most violently exactly when a company is growing fast and paying for that growth upfront (Chapters 1, 4). **Avoidance:** watch the **cash** curve, not the **revenue** curve.

Common pitfall These four are not independent — they **compound**. A founder who misprices (1) misreads their unit economics (2), over-hires against the resulting phantom revenue (3), and mistakes the doomed growth for health (4) until the bank balance betrays all four at once. The entire financial toolkit of this course exists to make each error **visible early**, while it is still cheap to fix.

11 The financial pitch and founder-side due diligence

11.1 Defending the model, not just presenting it

Everything in this book converges on a single test: a founder, in a room, defending a financial model to investors who are **trying to find the assumption that breaks it**. This is not a presentation; it is a **cross-examination**. The deliverable is not a beautiful deck but the **discipline of having an evidenced answer for every number** — and the honesty to mark which numbers are validated and which are still bets.

Definition 30 (the three differentiating slides) Beyond the standard narrative, three slides separate a serious financial pitch from a hopeful one — each the output of a chapter of this course:

1. **Unit economics** — CAC, LTV, LTV/CAC, payback, gross margin, with the **data** behind each (Chapter 3). Proves the business makes money on each customer.
2. **Runway** — current cash, net burn, months remaining, and the **milestone** this round buys before the next raise (Chapter 4). Proves the founder knows their deadline.
3. **Use of funds** — exactly how the new money converts into the next milestone, tied to the hiring plan and the model (Chapter 2). Proves the raise is a plan, not a wish.

11.2 Founder-side due diligence

Definition 31 (founder-side due diligence) Before pitching, the founder runs the **same** diagnostic an investor will run, on themselves: pressure-test every assumption, find the weakest number **first**, and prepare its defence (or fix it). The goal is that no investor question is a surprise — every hard question has already been asked internally and answered with evidence.

Worked example — an assumption under cross-examination. A founder projects LTV of €2,640 on **3% monthly churn** (Chapter 3). The investor's question is immediate: "Your company is 14 months old — how do you know churn over a 33-month average life is 3%?" The weak founder defends the number; the prepared founder **concedes its fragility** and shows the work: "Observed churn over our 14 months is 3.4%; I modelled 3% as a target and here is the cohort curve. At our **actual** 3.4%, LTV/CAC is 1.9× rather than 2.2×, and here is the retention initiative built to close the gap." The second answer survives diligence because it **anticipated** the attack — which is the entire point of running due diligence on yourself first.

Common pitfall **The fastest way to fail diligence is a model whose three statements do not reconcile** (Chapter 2). An investor who finds that P&L revenue, cash flow, and the cap table tell inconsistent stories stops trusting **every** number, not just the inconsistent one. Coherence is not a nicety; it is the precondition for being believed at all. The founder who has internalised the whole arc of this book — lifecycle, model, unit economics, burn, ecosystem, term sheet, cap table, waterfall — walks into that room with a model that ties, a deadline they own, and an answer for every assumption an investor can attack.

The arc of this course in one paragraph. A startup lives or dies on **cash**, not profit (Ch. 1). The founder forecasts that cash with a **bottoms-up model** whose three statements and cap table reconcile (Ch. 2), grounds it in **unit economics** that prove each customer pays for itself (Ch. 3), and watches it through **burn, runway, and break-even** (Ch. 4). To extend that runway they raise from investors whose **fund economics** dictate what they demand (Ch. 5), negotiating a **term sheet** whose preference, anti-dilution, and control clauses decide the real economics behind the headline valuation (Ch. 6). Each round **dilutes** the founder on the **cap table** (Ch. 7), and at exit a **waterfall** — preferences first, common last — converts ownership into actual euros (Ch. 8). Where equity is the wrong tool, **venture debt or revenue-based financing** fit a different cash profile (Ch. 9). The recurring **founder mistakes** (Ch. 10) are the shadows of these same disciplines, and the final test is **defending the model** under investor cross-examination (Ch. 11). Throughout, the founder reads every decision through two documents at once: the model that says whether the business works, and the cap table that says what they keep if it does.